The governing equation — its derivation, and its geometry

derivation — where it comes from

1. *Ramsey:* shielding is the field made by the electron currents the applied field stirs up.

2. reduce each source to an induced point dipole — its through-space field at the nucleus is dipolar.

3. that dipole field is the *second derivative* of the Coulomb potential $1/\rho$.

4. $\Rightarrow$ one tensor $\partial_a\partial_b(1/\rho)$ — the very same object as the dipolar coupling and the electric-field gradient.

$$\sigma_{ab}(i) \quad\approx\quad \sum_s I_s\, D_{ab}(\mathbf{r}_{is})$$

$$D_{ab} \;=\; \partial_a\partial_b\frac{1}{\rho} \;=\; \frac{3\hat{\rho}_a\hat{\rho}_b-\delta_{ab}}{\rho^3}$$

one object; the sum runs over nearby sources s

geometry — what it means

- **traceless & symmetric** — no isotropic part: pure anisotropy (the T_2 the thesis lives on).
- **eigenvalues** $+2/\rho^3$ along $\hat{\rho}$, $-1/\rho^3$ across (twice).
- **angular law** $(3\cos^2\theta-1)/\rho^3$.
- **magic-angle node** — vanishes at $\cos^2\theta=\frac{1}{3}$ ($\theta\approx 54.7°$).
- **radial** $1/\rho^3$ — steep, near-field.

the shape of D_{ab}